

010 Yellow Paper: Formal Specification of the v2 Proof System

010 contributors
010 v0.3.0 prototype

2026-09-30

Abstract

This document normatively specifies the 010 v0.3.0 proof system: symbols, domains, data structures with exact byte layouts, deterministic algorithms for proving and verification, the Fiat–Shamir transcript schedule, and test vectors. It is the companion to the whitepaper (protocol and theorems); in case of divergence between prose and code, the test suite decides, and this paper is amended.

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1 Symbols and domains

Symbol	Value	Definition
D	64	ring degree, $R_q = \mathbb{Z}_q[X]/(X^D + 1)$
q	1073750017	prime field modulus, $q = 2^{30} + 2^{13} + 1$
n, m	4, 8	commitment matrix shape over R_q
β, B	1, 64	witness norm bounds
S, S'	65536, 65472	mask / response bounds
w_c	16	challenge weight
τ	2	rounds
W_{64}	817737403	primitive 64-th root of unity mod q
ψ	187094501	128-th root with $\psi^2 = W_{64}$

Table 1: Frozen constants. All integers are unsigned unless stated.

Domain tags (single bytes): matrix expansion 0x41; round challenges 0x50; mask sampling 0x42; test witness sampler 0x77.

2 Data structures

Polynomials are 64 coefficients in $[0, q)$, coefficient-major, little-endian 32-bit words unless packed. A *centered* value is $c - q$ if $c > q/2$ else c .

Object	Bytes	Layout
t1 round commitment	1024	4 polys $\times$ LE32 coeffs
z response (unpacked)	2048	8 polys $\times$ LE32 coeffs
z response (packed)	1088	512×17 -bit offset-binary words, LSB-first
pack block	288	$u_0[4] 0[4] u_1[4] 0[4]$ then 272 zero bytes
chal block	64	master salt (32) then 32 zero bytes
proof	4577	version byte (2) then round ₀ , round ₁ , pack, chal

Table 2: Exact wire layout. Packed words are offset-binary $w = \bar{z} + 65536$ for centered $\bar{z}$; valid iff $64 \leq w \leq 131008$ (one check enforces canonicity and bound).

3 Algorithms

Definition 3.1 (Commit). $t = A \cdot w$ with negacyclic products via negative wrapped convolution: twist by ψ^j , size-64 NTT, pointwise multiply, inverse NTT, untwist. Normative result equals schoolbook $O(D^2)$ multiplication bitwise.

Definition 3.2 (Prove, per round j with salt s). For attempt $\in \{0, \dots, 7\}$: seed $\leftarrow H(s || j || \text{attempt})$; sample mask r uniform on $[-S, S]^{512}$ by rejection from the XOF stream; $t_1 \leftarrow A \cdot r$; $u \leftarrow \langle r, r \rangle$. Let h be the transcript output of Section 4; the challenge $c \leftarrow \text{Expand}(h)$ is sparse ternary with exactly 16 nonzero ± 1 entries. Respond $z \leftarrow r + c \star w$; accept iff $\|z\|_\infty \leq S'$.

Definition 3.3 (Verify). Parse version ($= 2$), enforce zero regions and packed-word ranges, then per round re-derive c and check $A \cdot z = t_1 + c \star y$ and $\|z\|_\infty \leq S'$. Reject on any failure (MalformedEncoding for encoding, VerificationFailed else).

4 Transcript schedule

FS absorbs, in order: `params` (D, q, β, B, S', τ , version), `A` (H of the flattened matrix), `y` (1024 B), `t`, `salt`, `round`, `t1` (1024 B), `u`. Absorption is $H(state \parallel label \parallel len_{16} \parallel data)$ (data over 1024 B is digested first); labels ≤ 64 B, data ≤ 2048 B. Challenges squeeze 32 B under tag `0x50`. The hash is BLAKE3-32 (spec-exact) with a documented counter-cascade XOF, not the official seek mode.

5 Test vectors

Matrix seed [11; 32], witness seed [22; 32], salt [33; 32] (full values: run `examples/vectors`):

- `y` (1024 B, LE32): begins `e406e318...`
- `t` = 4b010000 (LE32 of 331, the witness support weight).
- proof (4577 B, version byte 02), first 32 bytes `026d1d3f...`; SHA-256:

```
79095a012e44d5cbd784430bf449f8e668e72d25bf22
841189245615f59be624
```

BLAKE3 vectors: the 11 official $n \in \{0, 1, 2, 31, 32, 63, 64, 65, 1023, 1024, 1025\}$ cases with input $i \bmod 251$ (see `tests/kat_hash.rs`).

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